

MedVision AI Knowledge Base

Pneumonia

1. Overview

Pneumonia is an infection of the lung parenchyma that causes inflammation of the alveoli. The air sacs may fill with fluid or pus, reducing oxygen exchange and causing respiratory symptoms. It can be caused by bacteria, viruses, fungi, or aspiration. Severity ranges from mild outpatient illness to life-threatening disease requiring intensive care.

2. Causes

Bacterial: *Streptococcus pneumoniae*, *Haemophilus influenzae*, *Staphylococcus aureus*, atypical bacteria.

Viral: Influenza, RSV, SARS-CoV-2 and others.

Fungal: Opportunistic fungi, especially in immunocompromised patients.

Aspiration: Inhalation of food, gastric contents or saliva into the lungs.

3. Symptoms

Typical symptoms include fever, chills, productive or dry cough, shortness of breath, pleuritic chest pain, fatigue, rapid breathing, reduced oxygen saturation, and occasionally confusion in elderly patients.

4. Risk Factors

Advanced age, young children, smoking, COPD, emphysema, asthma, diabetes, chronic kidney disease, heart disease, cancer, immunosuppression, prolonged hospitalization, aspiration risk, alcohol misuse and malnutrition.

5. Chest X-ray Findings

Chest radiographs commonly demonstrate focal or diffuse consolidation, patchy infiltrates, air bronchograms, unilateral or bilateral opacities, lower-lobe predominance, multilobar disease, and pleural effusion. Early pneumonia may produce minimal radiographic abnormalities.

6. Diagnosis

Diagnosis integrates clinical history, physical examination, pulse oximetry, chest X-ray, complete blood count, inflammatory markers, sputum examination in selected cases, blood cultures in severe disease, and CT imaging when additional evaluation is required.

7. Differential Diagnosis

Important alternative diagnoses include pulmonary edema, atelectasis, tuberculosis, lung malignancy, pulmonary embolism, COVID-19 pneumonia, acute respiratory distress syndrome (ARDS), and organizing pneumonia.

8. Severity Classification

Severity may be classified as mild, moderate, severe, or critical based on respiratory compromise, oxygen requirements, imaging extent, laboratory findings, and validated clinical scores such as CURB-65 or PSI.

9. Treatment

Bacterial pneumonia is managed with guideline-directed antibiotics. Viral pneumonia usually requires supportive treatment, with antivirals when indicated. Severe disease may require oxygen therapy, intravenous fluids, non-invasive ventilation, mechanical ventilation, and intensive care.

10. Complications

Potential complications include respiratory failure, sepsis, pleural effusion, empyema, lung abscess, bacteremia, ARDS, multi-organ dysfunction, and death if untreated or severe.

11. Prevention

Preventive strategies include pneumococcal and influenza vaccination, smoking cessation, hand hygiene, respiratory etiquette, management of chronic illnesses, healthy nutrition, and early medical assessment for persistent respiratory symptoms.

12. Prognosis

Most healthy individuals recover completely with timely treatment. Prognosis depends on age, comorbidities, immune status, disease severity, and speed of intervention. Severe pneumonia carries a higher risk of complications and mortality.

13. Patient Explanation

Pneumonia means that part of your lung has become infected. The infection causes the tiny air sacs to fill with fluid instead of air, making breathing more difficult. Most patients improve with appropriate treatment, but some require hospital care depending on the severity of the infection.

14. AI Interpretation

An AI model detecting pneumonia should be considered a clinical decision-support tool. Review the confidence score, inspect highlighted regions, correlate imaging with symptoms, examination findings, and laboratory tests, and confirm the diagnosis with a qualified radiologist or physician.

15. Quick Facts

Organ: Lung

Imaging: Chest X-ray

Typical Finding: Consolidation

Common Symptom: Cough with fever

Possible Complication: Respiratory failure

Clinical Correlation Required: Yes

16. References

World Health Organization (WHO) – Pneumonia guidance.

MedlinePlus (U.S. National Library of Medicine) – Pneumonia overview.

Merck Manual Professional Edition – Pneumonia.

British Thoracic Society Guidelines – Community-acquired pneumonia.

American Thoracic Society / Infectious Diseases Society of America (ATS/IDSA) – Clinical practice guidelines.

Radiology Assistant and standard radiology textbooks for imaging features.